CHAPTER-2.

CHANGE OF AXES.

2.1: Transformation of coordinates.

The coordinates of a point is determined with respect to a particular system of axes and particular origin. If the system of axes changed the coordinates of the point also changed.

The system of axes can be changed in the following three ways;

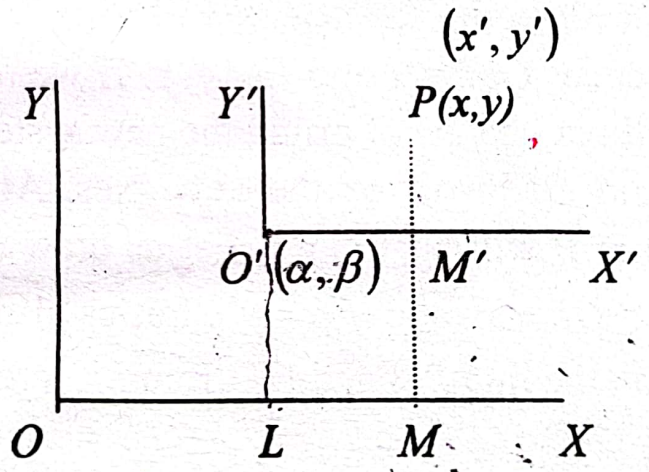
(i). By shifting the origin but the new axes remaining parallel to the original axes.

(ii). By rotating the axes through some angle without change of origin.

(iii). By shifting the origin as well as rotating the axes.

The above process of changing the coordinates of a point is called transformation of coordinates.

2.2: Shifting of origin.(The new axes remaining parallel to the original axes).



Let us consider OX and OY be the original axes of x and y respectively. Let $P(x,y)$ be a point in the XOY axes system. Let us draw a perpendicular PM on the axis OX. Here $OM = x$ and $PM = y$. If we transfer the origin at $O'(\alpha, \beta)$ referred to the original axes, the new axes will be $O'X'$ and $O'Y'$ which is parallel to the original axes. The coordinates $P(x,y)$ will be $P(x', y')$. Let us draw a perpendicular PM' to the axis $O'X'$. Here $O'M' = x'$ and $PM' = y'$

Here $OM = O'M' + OL = x' + \alpha$

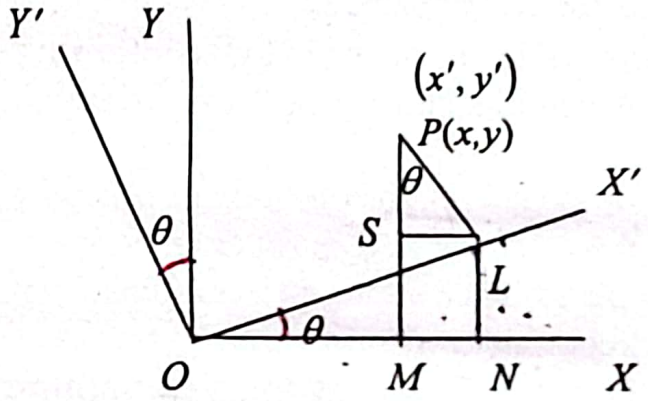
$$\therefore x = x' + \alpha$$

Here $PM = PM' + M'M = y' + \beta$ $[\because MM' = O'L = \beta]$

$\therefore y = y' + \beta$

Hence the relation between original and new coordinates of P are given by $x = x' + \alpha$ and $y = y' + \beta$

2.3: Rotation of axes without change of origin.



Let us consider OX and OY be the original axes of x and y. Let $P(x,y)$ be any point of on that system. Let us draw a perpendicular PM on the axis OX. Hence $OM = x$ and $PM = y$. If we rotate the axes through an angle θ without change of origin the new system will be $X'OY'$. So the line OX' and OY' will be the new axes. After rotating the axes the coordinates of P will be (x', y'). Let us draw a perpendicular PL to the axis OX'. $\therefore OL = x'$ and $PL = y'$. Let us also draw the perpendiculars LS and LN on the line PM and OX from L. Here $LS = MN$.

Here $x = OM = ON - MN = ON - LS = x'\cos\theta - y'\sin\theta$

From the triangle ΔOLN; $\cos\theta = \dfrac{ON}{OL} = \dfrac{ON}{x'}$; $\therefore ON = x'\cos\theta$

From the triangle ΔPLS; $\sin\theta = \dfrac{LS}{PL} = \dfrac{LS}{y'}$; $\therefore LS = y'\sin\theta$

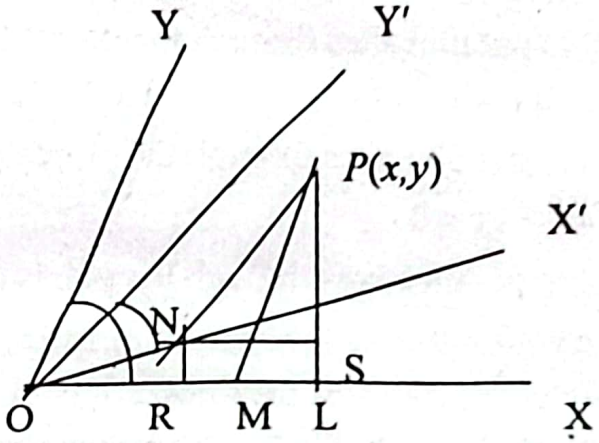
Again $y = PM = PS + SM = PS + LN = y'\cos\theta + x'\sin\theta$

Hence the relations between the original and new coordinates of the point P are given by $x = x'\cos\theta - y'\sin\theta$ and $y = x'\sin\theta + y'\cos\theta$

Solving for x' and y', we get

$$x' = x\cos\theta + y\sin\theta \text{ and } y' = y\cos\theta - x\sin\theta$$

2.4: Transition from one set of oblique axes to another with the same origin.



Let OX and OY be the original axes of x and y inclined at an angle ω. Let OX' and OY' be the new axes inclined at an angle ω'. Let θ be the angle between OX and OX' and ϕ be the angle between OX and OY'. Let $P(x,y)$ be any point referred to the original axes. This point will be (x, y') referred to the new axes. Let us draw the lines PM and PN parallel to OY and OY' axes.

$$\therefore OM = x;\ PM = y \text{ and } ON = x';\ PN = y'$$

Let us draw the perpendiculars PL, NR, NS from P,N on the lines OX, and PL. Since ϕ be the angle between OX and OY' then $\phi = \omega' + \theta$.

Then from the figure , we get

$$PM\sin\omega = PL = PS + SL = PS + NR = PN\sin\phi + ON\sin\theta$$

or, $$y\sin\omega = x'\sin\theta + y'\sin\phi$$

$$\therefore\ y = x'\frac{\sin\theta}{\sin\omega} + y'\frac{\sin\phi}{\sin\omega};\ \text{----(1)}$$

Since OX' and OY' are inclined to OY at angles $\omega - \theta$ and $\omega - \phi$ then

$$x\sin\omega = x'\sin(\omega - \theta) + y'\sin(\omega - \phi)$$

$$\therefore x = x'\frac{\sin(\omega - \theta)}{\sin\omega} + y'\frac{\sin(\omega - \phi)}{\sin\omega};\ \text{----(2)}$$

Here (1) and (2) are the relations between the coordinates of a point of original and new axes.

EXERCISE-2.

Change of Origin.

1. Transform to parallel axes through the points (1,–2), the equations

(*i*). $y^2 - 4x + 4y + 8 = 0$. (*ii*). $x^2 - 3y^2 + 4x + 6y = 0$

2. Transform to parallel axes through the points (1,–2), the equations $2x^2 + y^2 - 4x + 4y = 0$ **[DU-82].**

3. Transform to parallel axes through the points (3, 1), the equations $x^2 + 2y^2 - 6x + 7 = 0$

4. Prove that the equations of the lines $x + y = 1$ and $x - y = 1$ can be written as $X + Y = 0$ and $X - Y = 0$ by changing the origin to a suitable point.

5. By transferring to parallel axes through a properly chosen point, prove that the equation $x^2 + xy + y^2 + x + 5y + 7 = 0$ can be reduced to a homogeneous 2nd degree equation.

6. On shifting the origin to a point P, the axes remaining parallel to the original axes the equation $ax + by + c = 0$ is transformed to $ax + by + c + \lambda = 0$. Show that the locus of P is the line $ax + by = \lambda$.

7. By transferring the origin to a suitable point reduced the following equation $12x^2 + 2y^2 - 10xy + 11x - 5y + 2 = 0$ to a homogeneous form. **[RU-85]**

8. Remove the x and y terms of the following equations.

(*i*). $x^2 + 2y^2 - 4x + 6y - 8 = 0$ (*ii*) $3x^2 + 3y^2 - 12x + 12y - 1 = 0$

(*iii*). $2xy - x - y + 4 = 0$

Rotation of Axes.

9. Transform to axes inclined at 45^0 to the original axes, the equations

(*i*). $17x^2 - 16xy + 17y^2 = 225$

(*ii*). $x^2 - y^2 - 2\sqrt{2}x - 10\sqrt{2}y + 2 = 0$

(*iii*). $x^2 - y^2 = a^2$. (*iv*). $3x^2 + 3y^2 + 2xy - 2 = 0$

10. Remove the x, y, xy terms of the following equations

(i). $9x^2 + 24xy + 2y^2 - 6x + 20y + 41 = 0$

(ii). $17x^2 + 18xy - 7y^2 - 16x - 32y - 18 = 0$

(iii). $9x^2 + 15xy + y^2 + 12x - 11y - 15 = 0$

11. Remove the xy term of the following equations

(i). $5x^2 - 4\sqrt{3}xy + 9y^2 = 10$ (ii). $9x^2 - 2\sqrt{3}xy + 7y^2 = 10$

12. Find the angle through which the axes may be turned so that the equation $ax + by + c = 0$ may be reduced to $x = k$ and determined the value of k.

13. Find the angle through which the axes may be turned so that the equation $\sqrt{3}x + y - 4 = 0$ may be reduced to $x = k$ and determined the value of k.

14. Find the angle through which the axes may be turned so that the equation $x^2 - y^2 = a^2$ may be reduced to $xy = k^2$ and determined the value of k^2.

15. Transform the equation $14x^2 - 4xy + 11y^2 - 36x + 48y + 41 = 0$ to rectangular axes through the point $(1,-2)$ inclined at an angle $\tan^{-1}\left(\frac{-1}{2}\right)$ to the original axes.

16. Transform the equation $11x^2 + 24xy + 4y^2 - 20x - 40y - 5 = 0$ to rectangular axes through the point $(2,-1)$ inclined at an angle $\tan^{-1}\left(\frac{-4}{3}\right)$ to the original axes.

17. Determine the angle through which the axes must be rotated to remove the xy term in the following equations; (also transform the equation)

(i). $7x^2 - 6\sqrt{3}xy + 13y^2 = 16$ (ii). $9x^2 - 2\sqrt{3}xy + 7y^2 = 0$

18. Transform the equation $(ax + by + c)(bx - ay + d) = a^2 + b^2$ to new axes of x and y whose equations are the mutually perpendicular lines $ax + by + c = 0$, $bx - ay + d = 0$ respectively.

19. Transform the equation

56. Through what angle the axes be rotated to reduce the equation $3x^2 + 4xy - y^2 = 1$ in the form of $ax^2 + by^2 = 1$. Find also the value of a and b. Ans. $\theta = \frac{\pi}{8}$; $a = 1 + 2\sqrt{2}$; $b = 1 - 2\sqrt{2}$

57. Find the angle through which the axes must be turned so that the expression $ax^2 + 2hxy + by^2$ may become an expression in which the term xy is absent. Ans: $(1/2)\tan^{-1}(2h/(a-b))$.

SOLUTION OF EXERCISE-2.

Change of origin.

1.(i). Let $P(x, y)$ be a moving point on the curve $y^2 - 4x + 4y + 8 = 0$.

If we transfer the origin at $(1,-2)$, the coordinates will be (x', y').

$\therefore x = x' + 1$ and $y = y' - 2$

The transformed equation will be

$$(y' - 2)^2 - 4(x' + 1) + 4(y' - 2) + 8 = 0$$

or, $y'^2 - 4y' + 4 - 4x' - 4 + 4y' - 8 + 8 = 0$

or, $y'^2 = 4x'$

Hence the locus of (x', y') is $y^2 = 4x$ (Ans).

(ii). Let $P(x, y)$ be a moving point on the curve $x^2 - 3y^2 + 4x + 6y = 0$.

If we transfer the origin at $(1,-2)$, the coordinates will be (x', y').

$\therefore x = x' + 1$ and $y = y' - 2$

The transformed equation will be

$$(x' + 1)^2 - 3(y' - 2)^2 + 4(x' + 1) + 6(y' - 2) = 0$$

or, $x'^2 - 3y'^2 + 6x' + 18y' - 19 = 0$

Hence the locus of (x', y') is $x^2 - 3y^2 + 6x + 18y - 19 = 0$ **(Proved).**

2. If we transfer the origin at $(1,-2)$, the point $P(x, y)$ will be (x', y').

$\therefore x = x' + 1$, and $y = y' - 2$,

The transformed equation is

$$2(x' + 1)^2 + (y' - 2)^2 - 4(x' + 1) + 4(y' - 2) = 0$$

or, $2x'^2 + y'^2 = 6$

Hence the locus of (x', y') is $2x^2 + y^2 = 6$. **(Ans).**

3. Let $P(x, y)$ be a moving point. If we transfer the origin to the point $(3,1)$, then the coordinates $P(x, y)$ will be (x', y').

$$\therefore x = x' + 3, \quad y = y' + 1.$$

The transformed equation will be

$$(x' + 3)^2 + 2(y' + 1)^2 - 6(x' + 3) + 7 = 0$$

or, $x'^2 + 2y'^2 + 4y' = 0$

Hence the locus of (x', y') is $x^2 + 2y^2 + 4y = 0$ **(Ans).**

4. Let $P(x, y)$ be a variable point. If we transfer the origin at (h, k), the co-ordinates will be (X, Y).

$\therefore x = X + h$ and $y = Y + k$

The 1st equation will be

$X + h + Y + k - 1 = 0$ or, $X + Y + (h + k - 1) = 0$, ---(1)

The 2nd equation will be

$X + h - Y - k - 1 = 0$ or, $X - Y + (h - k - 1) = 0$, ---(2)

For making homogeneous (1) and (2), we are put

$h + k - 1 = 0$, ---(3)

and $h - k - 1 = 0$, ---(4)

Solving (3) and (4), we get, $h = 1$ $k = 0$

If we transfer the origin at $(1,0)$, the given equations will be

$X + Y = 0$ and $X - Y = 0$.

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5. If we transfer the origin at (h,k) the coordinates of the new axes system will be $(x+h, y+k)$. The given equation will be transformed as

$$(x+h)^2+(x+h)(y+k)+(y+k)^2+(x+h)+5(y+k)+7=0$$

or, $x^2+xy+y^2+(2h+k+1)x+(h+2k+5)y+$

$$+\left(h^2+hk+k^2+h+5k+7\right)=0$$

If the equation contains only 2nd degree terms then

$$2h+k+1=0$$

and $$h+2k+5=0$$

Solving, we get $$\frac{h}{5-2}=\frac{k}{1-10}=\frac{1}{4-1}$$

or, $$\frac{h}{3}=\frac{k}{-9}=\frac{1}{3};\quad \therefore h=1 \text{ and } k=-3$$

The transformed equation will be

$$x^2+xy+y^2+(1-3+9+1-15+7)=0$$

or, $$x^2+xy+y^2=0$$

If we transfer the origin at $(1,-3)$ the given equation will be transformed as homogeneous.

6. Let $P(\alpha,\beta)$ be a point. If we change the origin at P the coordinates of the original axis will be $(x+\alpha, y+\beta)$. The given equation will be

$$a(x+\alpha)+b(y+\beta)+c=0$$

or, $$ax+by+(c+a\alpha+b\beta)=0, \text{-----}(1)$$

Since (1) and $ax+by+\mathfrak{c}+\lambda=0$ are the same line then

$$c+\lambda=c+a\alpha+b\beta$$

$$\therefore a\alpha+b\beta=\lambda$$

Hence the locus of (α,β) is $ax+by=\lambda$ (Proved).

7. If we transfer the origin at (h,k) the coordinates of the new axes system will be $(x+h, y+k)$. The given equation will be transformed as

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$$12(x+h)^2+2(y+k)^2-10(x+h)(y+k)+11(x+h)-5(y+k)+2=0$$

$$\text{or, } 12x^2+2y^2-10xy+(24h-10k+11)x+(-10h+4k-5)y+$$
$$+(12h^2+2k^2-10hk+11h-5k+2)=0$$

If the equation contains only 2nd degree terms then

$$24h-10k+11=0$$

and $$-10h+4k-5=0$$

Solving, we get $$\frac{h}{50-44}=\frac{-k}{-120+110}=\frac{1}{96-100}$$

$$\text{or, } \frac{h}{6}=\frac{k}{10}=\frac{-1}{4};\ \therefore h=-\frac{3}{2} \text{ and } k=-\frac{5}{2}$$

The transformed equation will be

$$12x^2+2y^2-10xy+\left(\frac{108}{4}+\frac{50}{4}-\frac{150}{4}-\frac{33}{2}+\frac{25}{2}+2\right)=0$$

or, $12x^2+2y^2-10xy=0$ which is homogeneous.

If we transfer the origin at $\left(-\frac{3}{2}.-\frac{5}{2}\right)$ the given equation will be transformed as homogeneous.

8. (i). If we remove the terms of x and y from the given equation then we are to transfer the origin at (h,k). Then the old coordinates will be $(x+h, y+k)$. The given equation will be

$$(x+h)^2+2(y+k)^2-4(x+h)+6(y+k)-8=0$$

$$\text{or, } x^2+2y^2+(2h-4)x+(4k+6)y+(h^2+2k^2-4h+6k-8)=0$$

Since the terms of x and y are vanished then

$$2h-4=0 \text{ and } 4k+6=0;\ \therefore\ h=2 \text{ and } k=-\frac{3}{2}$$

The transformed equation will be

$$x^2+2y^2+\left(4+\frac{9}{2}-8-9-8\right)=0$$

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or, $2x^2+4y^2=33$ (Ans.)

8. (ii). If we remove the terms of x and y from the given equation then we are to transfer the origin at (h,k). Then the old coordinates will be $(x+h, y+k)$. The given equation will be

$$3(x+h)^2+3(y+k)^2-12(x+h)+12(y+k)-1=0$$

or, $3x^2+3y^2+(6h-12)x+(6k+12)y+(3h^2+3k^2-12h+12k-1)=0$

Since the terms of x and y are vanished then

$6h-12=0$ and $6k+12=0$; $\therefore\ h=2$ and $k=-2$

The transformed equation will be

$$3x^2+3y^2+(12+12-24-24-1)=0$$

or, $3x^2+3y^2=25$ (Ans.)

8. (iii). If we remove the terms of x and y from the given equation then we are to transfer the origin at (h,k). Then the old coordinates will be $(x+h, y+k)$. The given equation will be

$$2(x+h)(y+k)-(x+h)-(y+k)+4=0$$

or, $2xy+(2k-1)x+(2h-1)y+(2hk-h-k+4)=0$

If we remove the terms of x and y then

$$2h-1=0 \text{ and } 2k-1=0;\ \therefore h=\frac{1}{2},\ k=\frac{1}{2}$$

The transformed equation will be

$$2xy+\left(2.\frac{1}{2}.\frac{1}{2}-\frac{1}{2}-\frac{1}{2}+4\right)=0$$

or, $.4xy+7=0$ (Ans).

9(i). If we rotate the axes by an angle 45^0 without changing the origin the coordinates of a variable point will be

$$x\cos 45^0-y\sin 45^0;\ x\sin 45^0+y\cos 45^0$$

or, $\frac{1}{\sqrt{2}}(x-y);\ \frac{1}{\sqrt{2}}(x+y)$

The transformed equation is

$$17\left[\frac{(x-y)}{\sqrt{2}}\right]^2 - 16\left[\frac{(x-y)}{\sqrt{2}}\right]\left[\frac{(x+y)}{\sqrt{2}}\right] + 17\left[\frac{(x+y)}{\sqrt{2}}\right]^2 = 225$$

or, $17(x-y)^2 - 16(x-y)(x+y) + 17(x+y)^2 = 450$

or, $18x^2 + 50y^2 = 450$

or, $9x^2 + 25y^2 = 225$ (Ans.)

9(ii). If we rotate the axes by an angle 45^0 without changing the origin the coordinates of a variable point will be

$x\cos 45^0 - y\sin 45^0;\ x\sin 45^0 + y\cos 45^0$

or, $\frac{1}{\sqrt{2}}(x-y);\ \frac{1}{\sqrt{2}}(x+y)$

The transformed equation is

$$\left[\frac{1}{\sqrt{2}}(x-y)\right]^2 - \left[\frac{1}{\sqrt{2}}(x+y)\right]^2 - 2(x-y) - 10(x+y) + 2 = 0$$

or, $(x-y)^2 - (x+y)^2 - 4(x-y) - 20(x+y) + 4 = 0$

or, $-4xy - 24x - 16y + 4 = 0$

or, $xy + 6x + 4y = 1$ (Ans.)

9(iii). If we rotate the axes by an angle 45^0 without changing the origin the coordinates of a variable point will be

$x\cos 45^0 - y\sin 45^0;\ x\sin 45^0 + y\cos 45^0$

or, $\frac{1}{\sqrt{2}}(x-y);\ \frac{1}{\sqrt{2}}(x+y)$

The transformed equation is

$$\left[\frac{1}{\sqrt{2}}(x-y)\right]^2 - \left[\frac{1}{\sqrt{2}}(x+y)\right]^2 = a^2$$

or, $(x-y)^2 - (x+y)^2 = 2a^2$

or, $-4xy = 2a^2$

or, $2xy + a^2 = 0$ (Ans.)

9(iv). If we rotate the axes by an angle 45^0 without changing the origin the coordinates of a variable point will be

$x\cos 45^0 - y\sin 45^0;\ x\sin 45^0 + y\cos 45^0$ or, $\frac{1}{\sqrt{2}}(x-y);\ \frac{1}{\sqrt{2}}(x+y)$

The transformed equation is

$$3\left[\frac{1}{\sqrt{2}}(x-y)\right]^2 + 3\left[\frac{1}{\sqrt{2}}(x+y)\right]^2 + (x-y)(x+y) - 2 = 0$$

or, $3(x-y)^2 + 3(x+y)^2 + 2(x^2-y^2) - 4 = 0$

or, $8x^2 + 4y^2 = 4$ or, $2x^2 + y^2 = 1$ (Ans.)

10(i). If we change the origin at (h,k) the new coordinates of a variable point will be $(x+h, y+k)$. Then the transformed equation is

$$9(x+h)^2 + 24(x+h)(y+k) + 2(y+k)^2 - 6(x+h) + 20(y+k) + 41 = 0$$

or, $9x^2 + 24xy + 2y^2 + (18h + 24k - 6)x + (24h + 4k + 20)y +$

$$+ (9h^2 + 24hk + 2k^2 - 6h + 20k + 41) = 0$$

If we remove the terms of x and y, we get

$18h + 24k - 6 = 0$; i.e, $3h + 4k - 1 = 0$

and $24h + 4k + 20 = 0$; i.e, $6h + k + 5 = 0$

Solving, we get $\frac{h}{20+1} = \frac{k}{-6-15} = \frac{1}{3-24}$ $\therefore h = -1$ and $k = 1$

When the given equation is transformed to parallel axes through the point $(-1,1)$, it becomes $9x^2 + 24xy + 2y^2 + 54 = 0$; -----(1) whose 1st degree terms vanished.

If we apply the properties of invariant to remove the xy term from (1), we get $ax^2 + by^2 + 54 = 0$ ----(2)

where $a + b = 9 + 2 = 11$ ----(3) and $ab = 9\times 2 - 12^2 = -126$ ---(4)

Solving (3) and (4), we get $a = 18$ and $b = -7$

Hence the required equation is $18x^2 - 7y^2 + 54 = 0$ **(Ans).**

10(ii). If we change the origin at (h,k) the new coordinates of a variable

point will be $(x+h, y+k)$. Then the transformed equation is

$$17(x+h)^2+18(x+h)(y+k)-7(y+k)^2-16(x+h)-32(y+k)-18=0$$

$$17x^2+18xy-7y^2+2(17h+9k-8)x+2(9h-7k-16)y+$$

$$+17h^2+18hk-7k^2-16h-32k-18=0;----(1)$$

If we remove the terms of x and y then

$$17h+9k-8=0;----(2) \quad \text{and} \quad 9h-7k-16=0;----(3)$$

Solving (2) and (3), we get $h=1$ and $k=-1$

When the given equation is transformed to parallel axes through the point $(1,-1)$, it becomes

$17x^2+18xy-7y^2=10;-----(4)$ whose 1st degree terms vanished.

If we apply the properties of invariant to remove the xy term from (4), we get $ax^2+by^2=10----(5)$; where $a+b=17-7=10$ ----(6)

and $ab=17\times-7-9^2=-200---(7)$

Solving (6) and (7), we get $a=20$ and $b=-10$

Hence the required equation is

$$20x^2-10y^2=10 \text{ ; or } 2x^2-y^2=1 \textbf{ (Ans).}$$

10(iii). If we change the origin at (h,k) the new coordinates of a variable point will be $(x+h, y+k)$. Then the transformed equation is

$$9(x+h)^2+15(x+h)(y+k)+(y+k)^2+12(x+h)-11(y+k)-15=0$$

$$9x^2+15xy+y^2+(18h+15k+12)x+(15h+2k-11)y+$$

$$+9h^2+15hk+k^2+12h-11k-15=0;----(1)$$

If we remove the terms of x and y then we get

$$18h+15k+12=0;----(2) \quad \text{and} \quad 15h+2k-11=0;----(3)$$

Solving (2) and (3), we get $h=1$ and $k=-2$

After transferring the origin at $(1,-2)$ the given equation becomes

$$9x^2+15xy+y^2+2=0;----(4)$$

If we remove the xy term then by invariant property, we get

$$ax^2+by^2+2=0,----(5)$$

when $a+b=9+1=10,---(6)$ and $ab=9\times1-\left(\frac{15}{2}\right)^2=\frac{-189}{4},---(7)$

Solving (6) and (7), we get $a=\frac{27}{2}$ and $b=\frac{-7}{2}$

The required equation becomes

$$\frac{27}{2}x^2-\frac{7}{2}y^2+2=0 \quad \text{or, } 27x^2-7y^2+4=0 \quad \textbf{(Ans).}$$

11. (i) The given equation is $5x^2-4\sqrt{3}xy+9y^2=10$. If we remove the term of xy, then by invariant property, we get

$$ax^2+by^2=10;----(1)$$

where $a+b=5+9=14,---(2)$ and $ab=5\times9-\left(-2\sqrt{3}\right)^2=33;---(3)$

Here $b-a=\sqrt{(a+b)^2-4ab}=\sqrt{196-132}=8,---(4)$

Solving (3) and (4), we get $a=3$ and $b=11$

The required equation is $3x^2+11y^2=10$ **(Ans).**

11. (ii) The given equation is $9x^2-2\sqrt{3}xy+7y^2=10$. If we remove the term of xy, then by invariant property we get

$$ax^2+by^2=10;----(1)$$

where $a+b=9+7=16,---(2)$ and $ab=9\times7-\left(-\sqrt{3}\right)^2=60;---(3)$

Here $a-b=\sqrt{(a+b)^2-4ab}=\sqrt{256-240}=4,---(4)$

Solving (3) and (4), we get $a=10$ and $b=6$

The required equation is $10x^2+6y^2=10,$ or $5x^2+3y^2=5$ **(Ans).**

12. If we rotate the axes at an angle θ without change of origin then the given equation will be $x=k$. After rotating the axes the variable coordinates will be $(x\cos\theta-y\sin\theta, x\sin\theta+y\cos\theta)$. After rotating the axes the equation will be

$$a(x\cos\theta-y\sin\theta)+b(x\sin\theta+y\cos\theta)+c=0$$

or, $(a\cos\theta+b\sin\theta)x+(b\cos\theta-a\sin\theta)y=-c$

or, $a_1x+b_1y=-c$

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where $a_1 = a\cos\theta + b\sin\theta;\quad b_1 = b\cos\theta - a\sin\theta$

Since $b_1 = 0$, then $b\cos\theta - a\sin\theta = 0,\ \therefore \theta = \tan^{-1}\left(\frac{b}{a}\right)$

So the equation will be $x = k$,

where $k = -\frac{c}{a_1} = -\frac{c}{a\cos\theta + b\sin\theta}$, Also, $b\cos\theta - a\sin\theta = 0$

$$\frac{\cos\theta}{a} = \frac{\sin\theta}{b} = \frac{\sqrt{\cos^2\theta + \sin^2\theta}}{\sqrt{a^2+b^2}} = \frac{1}{\sqrt{a^2+b^2}}$$

$$\therefore \cos\theta = \frac{a}{\sqrt{a^2+b^2}},\ \text{and}\ \sin\theta = \frac{b}{\sqrt{a^2+b^2}}$$

$$\therefore a\cos\theta + b\sin\theta = \frac{a^2+b^2}{\sqrt{a^2+b^2}} = \sqrt{a^2+b^2} \quad \therefore k = -\frac{c}{\sqrt{a^2+b^2}}$$

13. If we rotate the axes at an angle θ without change of origin then the given equation will be $x = k$. After rotating the axes the variable coordinates will be $(x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta)$. After rotating the axes the equation will be

$$\sqrt{3}(x\cos\theta - y\sin\theta) + (x\sin\theta + y\cos\theta) - 4 = 0$$

or, $(\sqrt{3}\cos\theta + \sin\theta)x + (\cos\theta - \sqrt{3}\sin\theta)y = 4$; or, $a_1x + b_1y = 4$

where $a_1 = \sqrt{3}\cos\theta + \sin\theta;\ b_1 = \cos\theta - \sqrt{3}\sin\theta$

Since $b_1 = 0$, then $\cos\theta - \sqrt{3}\sin\theta = 0,\ \therefore \theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 30^0$

So the equation will be $x = k$,

$$\text{where } k = \frac{4}{\sqrt{3}\cos 30^0 + \sin 30^0} = \frac{4}{\sqrt{3}\times\frac{\sqrt{3}}{2} + \frac{1}{2}} = \frac{4}{2} = 2\ .\ (\text{Ans.})$$

14. If we rotate the axes at an angle θ without change of origin then the given equation will be $xy = k^2$. After rotating the axes the variable coordinates will be $(x\cos\theta - y\sin\theta, x\sin\theta + y\cos\theta)$.

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After rotating the axes the equation will be

$$(x\cos\theta - y\sin\theta)^2 - (x\sin\theta + y\cos\theta)^2 = a^2$$

$$\text{or, } a_1x^2 + b_1xy + c_1y^2 = a^2$$

where $a_1 = \cos^2\theta - \sin^2\theta$; $c_1 = \sin^2\theta - \cos^2\theta$ and $b_1 = -4\cos\theta.\sin\theta$

Since $a_1 = c_1 = 0$, then $\cos^2\theta - \sin^2\theta = 0 \;\therefore\; \theta = \frac{\pi}{4}$

So the equation may be written as $b_1xy = a^2$; or $xy = k^2$

where $k^2 = -\frac{a^2}{4\cos 45^0.\sin 45^0} = -\frac{a^2}{2}$ (Ans).

15. If we the origin at $(1,-2)$, the given equation is transformed to

$$14(x+1)^2 - 4(x+1)(y-2) + 11(y-2)^2 - 36(x+1) + 48(y-2) + 41 = 0$$

$$\text{or, } 14x^2 - 4xy + 11y^2 = 25, \text{----(1).}$$

If the axes tern through an angle $\theta = \tan^{-1}\left(\frac{-1}{2}\right)$, or, $\tan\theta = -\frac{1}{2}$

then we have $\sin\theta = -\frac{1}{\sqrt{5}}$; and $\cos\theta = \frac{2}{\sqrt{5}}$.

$$\therefore x\cos\theta - y\sin\theta = \frac{1}{\sqrt{5}}(2x+y) \text{ and } x\sin\theta + y\cos\theta = \frac{1}{\sqrt{5}}(2y-x)$$

The equation becomes

$$14\left(\frac{2x+y}{\sqrt{5}}\right)^2 - 4\left(\frac{2x+y}{\sqrt{5}}\right)\left(\frac{2y-x}{\sqrt{5}}\right) + 11\left(\frac{2y-x}{\sqrt{5}}\right)^2 = 25$$

$$\text{or, } 3x^2 + 2y^2 = 5 \text{ (Ans.)}$$

16. If change the origin to $(2,-1)$, the given equation is transformed to

$$11(x+2)^2 + 24(x+2)(y-1) + 4(y-1)^2 - 20(x+2) - 40(y-1) - 5 = 0$$

$$\text{or, } 11x^2 + 24xy + 4y^2 = 5; \text{----(1)}$$

If the axes tern through an angle $\theta = \tan^{-1}\left(\frac{-4}{3}\right)$, or, $\tan\theta = -\frac{4}{3}$

then we have $\sin\theta = -\frac{4}{5}$; and $\cos\theta = \frac{3}{5}$.

$\therefore x\cos\theta - y\sin\theta = \frac{1}{5}(3x+4y)$ and $x\sin\theta + y\cos\theta = \frac{1}{5}(3y-4x)$

The equation becomes

$$11\left(\frac{3x+4y}{5}\right)^2 + 24\left(\frac{3x+4y}{5}\right)\left(\frac{3y-4x}{5}\right) + 4\left(\frac{3y-4x}{5}\right)^2 = 5$$

or, $x^2 - 4y^2 + 1 = 0$ (Ans.)

17(i). If we rotate the axes through an angle $\theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$ then the xy term vanished, where, $2h = -6\sqrt{3}$; $a = 7$ and $b = 13$

$\therefore \theta = \frac{1}{2}\tan^{-1}\left(\frac{-6\sqrt{3}}{7-13}\right) = \frac{1}{2}\tan^{-1}\left(\sqrt{3}\right) = 30^0$ (Ans.)

17(ii). If we rotate the axes through an angle $\theta = \frac{1}{2}\tan^{-1}\left(\frac{2h}{a-b}\right)$ then the xy term vanished, where, $2h = -2\sqrt{3}$; $a = 9$ and $b = 7$

$\therefore \theta = \frac{1}{2}\tan^{-1}\left(\frac{-2\sqrt{3}}{9-7}\right) = \frac{1}{2}\tan^{-1}\left(-\sqrt{3}\right) = 60^0$ (Ans.)

18. Let (x', y') be the coordinates of a variable point.

Since $ax + by + c = 0$ and $bx - ay + d = 0$ are the transformed axes of x and y then $x' = \frac{bx - ay + d}{\sqrt{a^2+b^2}}$, and $y' = \frac{ax + by + c}{\sqrt{a^2+b^2}}$ ----(1)

The given equation may be written as

$$(ax+by+c)(bx-ay+d) = a^2+b^2 \text{ or, } \left(\frac{ax+by+c}{\sqrt{a^2+b^2}}\right)\left(\frac{bx-ay+d}{\sqrt{a^2+b^2}}\right) = 1$$

Putting the value of x' and y', we get $x'y' = 1$

Hence the locus of (x', y') is $xy = 1$

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